

Agent Design Document

Title: Swarm Intelligence for Search and Rescue Operations using Decentralized Drone Agents

INTRODUCTION

This document presents the design of a decentralized multi-agent system for search and rescue operations using a swarm of autonomous drones. The system's main goal is to work together to explore and map an unknown disaster area, and to find victims, without needing a central controller. Each drone acts as an intelligent agent, using local sensing, communication, and decision-making based on utility to create a swarm that is both efficient and reliable.

1. PEAS ANALYSIS

1.1 Performance Measure:

- **Area coverage efficiency:**
The primary objective of the swarm is to maximize the percentage of the disaster zone explored within minimum time. Higher coverage in fewer steps indicates better swarm coordination.
- **Victim detection accuracy:**
The system should correctly detect and report victims present in the environment while minimizing false detections.
- **Collision avoidance:**
Drones must avoid collisions with obstacles and other drones to ensure safe operation.
- **Minimized exploration overlap:**
Repeated exploration of already mapped areas should be minimized to improve overall efficiency.
- **Robustness to agent failure:**
The system should continue operating effectively even if some drones fail or lose communication.

1.2 Environment:

- **Disaster zone:**
A partially known or unknown environment represented as a grid or continuous space containing obstacles, hazards, and victims.
- **Dynamic conditions:**
The environment may change due to movement of drones, discovery of new information, or simulated hazards.
- **Presence of multiple agents:**
Several autonomous drones operate simultaneously within the same environment.

1.3 Actuators:

- **Movement control:**
Each drone can move in multiple directions within the environment to explore and navigate.
- **Communication broadcast:**
Drones can transmit information such as explored areas and victim locations to nearby agents.
- **Hover and stop actions:**
Used for scanning areas or confirming victim detection.

1.4 Sensors:

- **Obstacle sensors:**
Detect nearby obstacles to prevent collisions.
- **Local area sensing:**
Identify whether surrounding areas have been explored or remain unexplored.
- **Victim detection sensor:**
A simulated sensor such as a thermal or visual sensor, identifies victims within a restricted range.

- **Neighbour detection:**
Identification of nearby drones that are within communication range.

2. ENVIRONMENT PROPERTIES

2.1 Observability (Fully Observable or Partially Observable):

- **Partially Observable:**
Each drone can perceive only a limited local area. The complete environment map is not known initially and is built incrementally through exploration and communication.

2.2 Determinism (Deterministic or Stochastic):

- **Stochastic:**
Uncertainty is introduced by sensor noise, communication delays, and dynamic agent interactions, thereby making outcomes non-deterministic.

2.3 Temporal Nature (Episodic or Sequential):

- **Sequential:**
Each action taken by a drone affects future states of the environment, such as explored regions and drone positions.

2.4 Dynamics (Static or Dynamic):

- **Dynamic:**
The environment continuously changes as drones move, share information, and detect victims.

2.5 State Space (Discrete or Continuous):

- **Discrete:**
The environment is modelled as a grid where exploration, movement, and communication occur in discrete steps.

2.6 Agency (Single-agent or multi-agent):

- **Multi-Agent:**
Multiple autonomous drone agents operate concurrently and cooperatively without a central controller.

3. ARCHITECTURE CHOICE AND JUSTIFICATION

The system adopts a utility-based decentralized agent architecture to enable efficient coordination among drones.

3.1 Choice of Utility-Based Agent:

A utility-based agent evaluates multiple possible actions and selects the action that **maximizes a utility function**. This utility function consists of **exploration gain, distance cost, collision risk, and overlap penalty**. This allows each drone to make intelligent trade-offs and adapt dynamically to changing conditions.

Utility function Definition:

The utility of an action a for a drone ' i ' is defined as:

$$U(a) = \alpha \cdot (\text{Exploration Gain}) - \beta \cdot (\text{Distance Cost}) - \gamma \cdot (\text{Collision Risk}) - \delta \cdot (\text{Overlap Penalty}) - \epsilon \cdot (\text{Battery cost})$$

Where:

- **Exploration Gain** → number of new unexplored cells.
- **Distance Cost** → movement effort.
- **Collision Risk** → proximity to obstacles/drones.
- **Overlap Penalty** → visiting an already explored areas.
- **Battery cost** → Represents the energy required to reach
- $\alpha, \beta, \gamma, \delta, \epsilon$ are controllable weights.

3.2 Rejection of Reflex-Based Agent:

A purely reflex-based agent reacts only to immediate sensor inputs and it lacks memory or foresight, leading to inefficient exploration and high overlap.

3.3 Rejection of Goal-Based Agent:

A goal-based agent can aim to explore unknown areas, it does not inherently optimize competing objectives such as safety, energy efficiency, and coordination with nearby drones.

3.4 Inter-Agent Communication:

Agents communicate using local peer-to-peer broadcasts within a limited range. Only essential information such as newly explored map cells, detected obstacles, victim locations, and short-term movement intentions are shared. The information is merged to the agent's local knowledge and is also used to update utility functions. Cooperation emerges only through decentralized communication and no agent has global knowledge.

4. INTERNAL AND EXTERNAL AGENT CHARACTERISTICS:

4.1 Internal Characteristics

- **Local map memory:**
Each drone maintains a partial map of explored and unexplored regions.
- **Utility evaluation mechanism:**
Actions are selected based on a utility function balancing exploration benefit and movement cost.
- **Belief state:**
The agent maintains beliefs about unknown areas, which are based on incomplete information.
- **Decision policy:**
Rules governing movement, communication, collision avoidance, and exploration priorities.

4.2 External Characteristics:

- **Physical environment interaction:**
The agent operates within a simulated disaster zone containing obstacles and victims.
- **Communication with neighbours:**
Information sharing occurs locally between nearby drones.
- **Autonomous execution:**
Each drone operates independently without human intervention or centralized control.

5. FRONTEND-BACKEND INTEGRATION

The system follows a **modular architecture** separating visualization and agent intelligence.

5.1 Frontend Design:

- **Environment visualization:**
Displays the disaster map, explored regions, identifies obstacles and victims, and indicates drone positions.
- **Performance metrics display:**
Provides information on area coverage percentage, time steps, and the number of victims detected.
- **Simulation control interface:**
Allows starting, pausing, or resetting the simulation.

5.2 Backend Design:

- **Environment engine:**
Manages map generation, obstacles, and victim placement.
- **Agent controller:**
Implements sensing, decision-making, movement, and communication logic for each drone.
- **Swarm co-ordination logic:**
Handles local communication and map merging between drones (agents).
- **Data logging module:**

Records metrics for evaluation and analysis.

5.3 Integration Mechanism:

The integration is achieved through an iterative simulation loop:

1. Backend updates the simulation state:
sensing → decision making → movement → communication.
2. The backend then produces an updated world state, which includes the positions of the drones, the regions they've explored, and any victims they've detected.
3. The frontend then renders the updated map and displays live metrics. This loop continues until a termination condition is met, such as full area coverage, all victims found, or a time limit being reached.

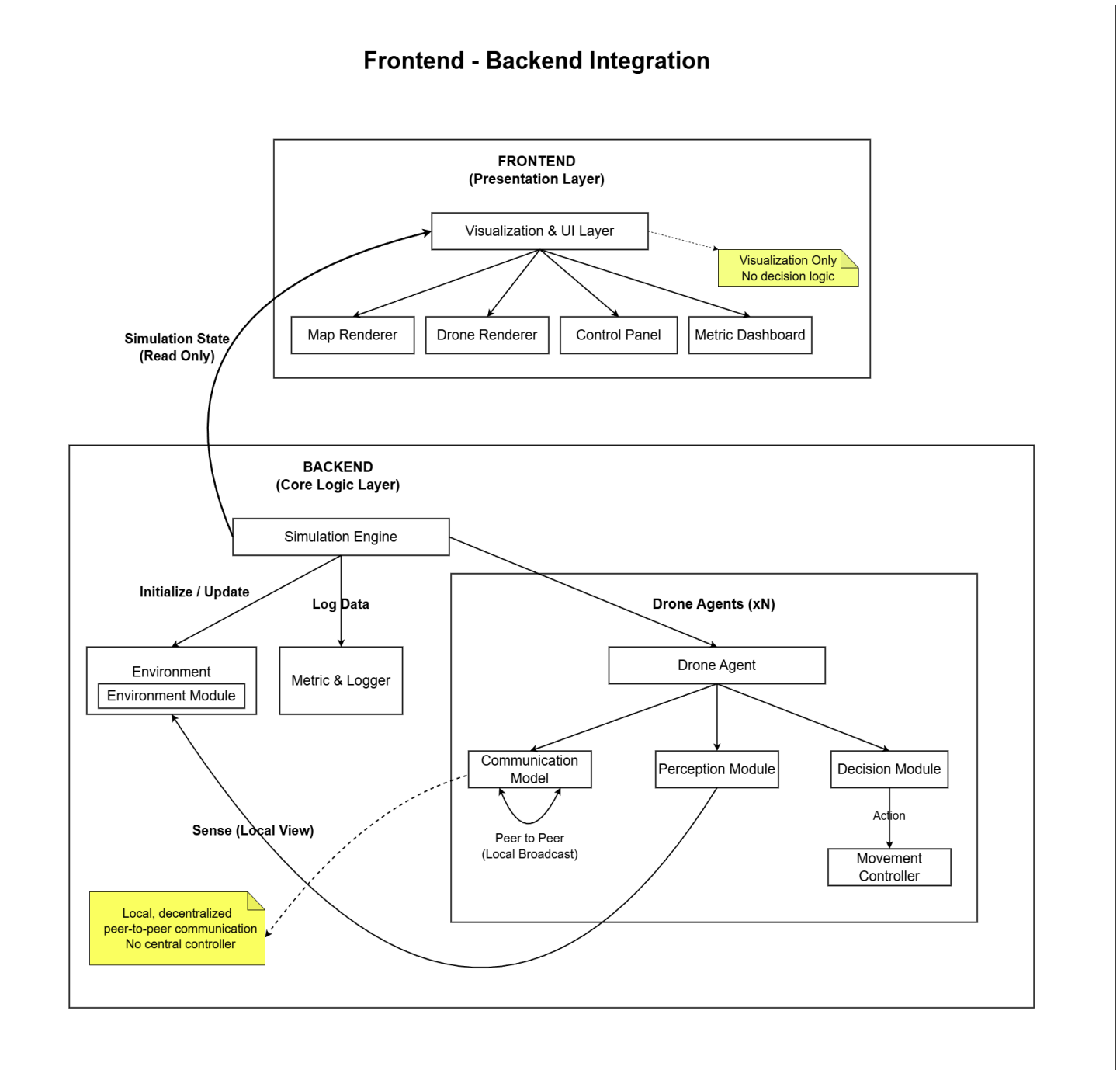


Figure 1: Frontend-Backend Integration Diagram

6. SYSTEM ASSUMPTIONS AND CONSTRAINTS

- All drones start with equal capabilities (homogeneous agents).
- Drones have limited battery life.
- Communication is local and unreliable.
- Victim detection sensors may produce false positives/negatives.
- The disaster environment is bounded and finite.

7. CONCLUSION

This document outlined the design of a decentralized swarm intelligence system for search and rescue using autonomous drones. By adopting a utility-based agent architecture, the system enables efficient exploration, victim detection, and coordination without a central controller. The proposed design addresses challenges such as partial observability, dynamic environments, and agent failures through local sensing and communication. Overall, the architecture provides a scalable and robust foundation for implementing and evaluating swarm-based search and rescue operations in future work.